Course Title	Basic Mechanical Engineering	**Course Code**	ME2001
Department	Department of Electrical Engineering (DEE)	**Campus**	Lahore
Knowledge Profile	Engineering Fundamentals (WK3)	**Credit Hrs.**	3
Knowledge Area	Professional Studies (KA07)	**Grading Scheme**	Relative
HEC Knowledge Area	Multi-Disciplinary Engg. Breadth (Elective)	**Applicable From**	Spring 2026
SDG	4 | Quality Education		
Pre-requisite(s)			

Course Objective	To build the fundamental concepts of engineering mechanics and to clearly present their application in the analysis of structures and planner motion of mechanical components.

No.	**Assigned Program Learning Outcome (PLO)**
02	An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

A = Assignment, Q = Quiz, M = Midterm, F = Final

No.	**Course Learning Outcome (CLO) Statements**	**Assessment Tools**	**Taxonomy Levels**	**PLO**
01	Calculate the moment of a force/couple using scalar and vector formulation.	Q1, A1, M1	C3	2
02	Apply the concept of mechanics to solve problems related to 2D static equilibrium analysis of a rigid body, dry friction, center of gravity, and moment of inertia.	Q2, A2, M2	C3	2
03	Analyze the kinematics and kinetics of rigid bodies undergoing translation, rotation, and general plane motion using Newton's laws of motion.	Q3, A3, F	C4	2
04	Evaluate the internal forces in the members of a loaded truss, beams, and frames to assess safety of the structure.	M2, F	C5	2

Text Book(s)	**Title**	Engineering Mechanics: Statics and Dynamic, 12th Ed
	Author	R. C. Hibbeler
	Publisher	Pearson Prentice Hall
Ref. Book(s)	**Title**	Statics and Mechanics of Materials,4th Ed
	Author	R. C. Hibbeler
	Publisher	Pearson Prentice Hall
	Title	Engineering Mechanics: Statics and Dynamics
	Author	Meriam, J. L. and Kraige, L. G
	Publisher	John & Wiley Sons

Week	Course Contents/Topics	Chapter*	CLO*
1	General Principles, Definitions and Scope of the Course in EE	1	1
2	Introduction to Vector Mechanics	2	1
3	Force Systems with Moments and Couple	4.1 to 4.6	1
4	Free Body Diagram and Equilibrium of Rigid Body in 2D	3.2, 5.1 to 5.3, 5.6	1, 2
5	Dry Friction and its Applications	8.1, 8.2	2
6	Centre of Gravity, Centroid, and Moment of Inertia	9.1, 10.1 to 10.4	2
7	Analysis of Beams and Frames – External and Internal Effects	7.1 to 7.3 (5.6, 5.7)	4
8	Structure Analysis: Method of Joints	6.1, 6.2	4
9	Structure Analysis: Zero-Force Members and Method of Sections	6.3, 6.4	4
10	Safety of the Structure, Factor of Safety Calculations	Lecture Notes	4
11	<u>Planar Kinematics of Rigid Body:</u> Different Types of Motion and Absolute Motion Analysis	16.1 to 16.4	3
12	Relative Motion Analysis: Velocity and Acceleration	16.5, 16.7	3
13	<u>Kinetics of a Particle: Force and Acceleration</u> Newton's Second Law of Motion and Equations of Motion in Rectangular and Cylindrical Coordinate Systems	13.1 to 13.6	3
14	<u>Planar Kinetics of Rigid Body: Force and Acceleration</u> Mass Moment of Inertia and Equations of Motion	17.1, 17.2	3
15	Equations of Motion: Rectilinear and Curvilinear Translation	17.3	3

***Reference book chapters are given in brackets**

Assessment Tools	Weightage
Quizzes (3), Assignments (3)	20.0%
Midterm (1+2)	30.0%
Final Exam	50.0%